P0 Walk-Forward Validation

Point-in-time backtest of price-derived factor blocks | 2018-11-01 to 2020-06-30 | monthly rebalance, 20 dates | generated 2026-08-20T17:29:38

What this report does and does not establish

This is not a validation of Model 5.0. It exercises only the price-derived blocks. Quality, growth and value together carry 70% of the production score and require point-in-time fundamentals that are not yet ingested, so none of them are tested here.

It answers one question from p0.md §7E: **how much of the model's behaviour is momentum?** A momentum-only ranking that performs close to the full model would mean the fundamental blocks are decoration. Read every number below as evidence about momentum and risk, not about Model 5.0.

Two further gaps: these ablations rank **market-wide** where production ranks within sector, and momentum-only omits RS_Sector_6M_Pct (20% of the production momentum block) because the bhavcopy archive carries no sector classification.

Integrity controls

Control	Status
Universe source	Archived NSE bhavcopy per session (point-in-time, includes delisted names)
Survivorship	3384 securities; 522 delisted retained in-sample
Identifier	ISIN-bridged Security_ID; 419 face-value changes linked
Execution	Signal on close of t, fill at open of next confirmed session
Corporate actions	659 split/bonus adjusted, 11580 dividends, 620 unadjustable events excluded
Delisting policy	haircut@0.50
Costs	effective-dated Indian charges + 10bp half-spread + sqrt impact at Rs 1.00 lakh/position
Eligibility	min turnover Rs 20.00 lakh/day, trading frequency >= 0.8, history >= 200 sessions, INE-class only (ETFs excluded)
Parameters	Frozen; no re-optimisation across the window

Eligible universe per rebalance

Metric	Value
Rebalance dates	20
Mean eligible	850
Min / max eligible	785 / 925
Mean considered before filters	1567

Rank information coefficient — Gross of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	14	0.0059	-0.0073	43%	-0.0629	0.1084
growth only	3M	14	-0.0065	-0.0148	29%	-0.0511	0.0742
growth only	6M	14	-0.0107	-0.0078	29%	-0.0484	0.0145
model 5	1M	20	0.0751	0.1007	70%	-0.3310	0.3328
model 5	3M	20	0.1304	0.1192	90%	-0.3084	0.4040
model 5	6M	20	0.1755	0.2048	85%	-0.1735	0.4272
model 5 gated	1M	20	0.0751	0.1007	70%	-0.3310	0.3328
model 5 gated	3M	20	0.1304	0.1192	90%	-0.3084	0.4040
model 5 gated	6M	20	0.1755	0.2048	85%	-0.1735	0.4272
momentum only	1M	20	0.0618	0.0761	70%	-0.2773	0.2910
momentum only	3M	20	0.1038	0.0764	90%	-0.2364	0.3872
momentum only	6M	20	0.1364	0.1702	85%	-0.1661	0.3481
quality only	1M	20	0.0670	0.0848	75%	-0.2668	0.2265
quality only	3M	20	0.0973	0.1054	85%	-0.2019	0.2931
quality only	6M	20	0.1309	0.1527	80%	-0.1055	0.3065
random ranking	1M	20	0.0044	-0.0012	40%	-0.0870	0.0642
random ranking	3M	20	0.0106	0.0209	65%	-0.0462	0.0597
random ranking	6M	20	0.0148	0.0173	65%	-0.0604	0.0695
simple quality momentum	1M	20	0.0595	0.0883	70%	-0.3245	0.3147
simple quality momentum	3M	20	0.1057	0.0744	90%	-0.3202	0.3937
simple quality momentum	6M	20	0.1460	0.1855	80%	-0.1778	0.3717
value only	1M	20	0.0232	0.0185	65%	-0.1080	0.1543
value only	3M	20	0.0443	0.0465	70%	-0.1096	0.1828
value only	6M	20	0.0531	0.0723	70%	-0.0688	0.1737

Portfolio and bucket evidence — Gross of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	-0.03%	-0.03%	+0.00%	-0.018	+0.00%
equal weight universe	3M	1.32%	1.32%	+0.00%	-0.236	+0.34%

equal weight universe	6M	5.71%	5.71%	+0.00%	-0.382	-0.89%
growth only	1M	-0.03%	-0.21%	-0.18%	-0.418	-0.04%
growth only	3M	1.32%	-0.56%	-1.89%	-0.224	-0.37%
growth only	6M	5.71%	0.12%	-5.58%	-0.648	-2.01%
model 5	1M	-0.03%	2.63%	+2.66%	0.782	+2.87%
model 5	3M	1.32%	9.99%	+8.66%	0.879	+7.86%
model 5	6M	5.71%	20.75%	+15.04%	0.758	+13.74%
model 5 gated	1M	-0.03%	2.63%	+2.66%	0.782	+2.87%
model 5 gated	3M	1.32%	9.99%	+8.66%	0.879	+7.86%
model 5 gated	6M	5.71%	20.75%	+15.04%	0.758	+13.74%
momentum only	1M	-0.03%	1.55%	+1.58%	0.733	+2.37%
momentum only	3M	1.32%	5.91%	+4.59%	0.806	+8.64%
momentum only	6M	5.71%	12.40%	+6.70%	0.503	+10.93%
quality only	1M	-0.03%	0.53%	+0.56%	0.721	+0.28%
quality only	3M	1.32%	3.11%	+1.79%	0.527	-0.21%
quality only	6M	5.71%	10.47%	+4.76%	0.612	+2.26%
random ranking	1M	-0.03%	0.65%	+0.68%	0.176	-0.01%
random ranking	3M	1.32%	1.34%	+0.02%	0.406	+0.58%
random ranking	6M	5.71%	5.07%	-0.64%	0.576	+2.77%
simple quality momentum	1M	-0.03%	1.40%	+1.42%	0.661	+1.78%
simple quality momentum	3M	1.32%	7.31%	+5.98%	0.867	+8.33%
simple quality momentum	6M	5.71%	14.98%	+9.28%	0.467	+11.70%
value only	1M	-0.03%	3.45%	+3.48%	0.491	+1.61%
value only	3M	1.32%	11.01%	+9.69%	0.697	+5.23%
value only	6M	5.71%	67.95%	+62.25%	0.770	+21.72%

Rank information coefficient — Net of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	14	0.0040	-0.0068	43%	-0.0648	0.1092
growth only	3M	14	-0.0076	-0.0151	29%	-0.0485	0.0768
growth only	6M	14	-0.0117	-0.0108	29%	-0.0506	0.0126
model 5	1M	20	0.1048	0.1354	75%	-0.3170	0.3537
model 5	3M	20	0.1468	0.1327	90%	-0.2977	0.4196

model 5	6M	20	0.1860	0.2169	85%	-0.1672	0.4345
model 5 gated	1M	20	0.1048	0.1354	75%	-0.3170	0.3537
model 5 gated	3M	20	0.1468	0.1327	90%	-0.2977	0.4196
model 5 gated	6M	20	0.1860	0.2169	85%	-0.1672	0.4345
momentum only	1M	20	0.0903	0.1146	75%	-0.2646	0.3154
momentum only	3M	20	0.1197	0.0901	90%	-0.2261	0.4039
momentum only	6M	20	0.1464	0.1839	85%	-0.1604	0.3533
quality only	1M	20	0.0802	0.0986	75%	-0.2577	0.2350
quality only	3M	20	0.1037	0.1119	85%	-0.1975	0.2948
quality only	6M	20	0.1354	0.1578	80%	-0.0994	0.3093
random ranking	1M	20	0.0044	0.0011	55%	-0.0991	0.0670
random ranking	3M	20	0.0106	0.0204	65%	-0.0464	0.0569
random ranking	6M	20	0.0149	0.0171	60%	-0.0596	0.0691
simple quality momentum	1M	20	0.0887	0.1295	70%	-0.3104	0.3363
simple quality momentum	3M	20	0.1220	0.0919	90%	-0.3091	0.4096
simple quality momentum	6M	20	0.1563	0.1989	85%	-0.1712	0.3778
value only	1M	20	0.0156	0.0037	50%	-0.1180	0.1381
value only	3M	20	0.0399	0.0415	70%	-0.1163	0.1764
value only	6M	20	0.0504	0.0687	70%	-0.0781	0.1703

Portfolio and bucket evidence — Net of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	-2.25%	-2.25%	+0.00%	0.164	+0.32%
equal weight universe	3M	-0.90%	-0.90%	+0.00%	-0.236	+0.67%
equal weight universe	6M	3.48%	3.48%	+0.00%	-0.382	-0.56%
growth only	1M	-2.25%	-2.90%	-0.65%	0.054	+0.08%
growth only	3M	-0.90%	-3.25%	-2.35%	-0.139	-0.25%
growth only	6M	3.48%	-2.57%	-6.05%	-0.661	-1.88%
model 5	1M	-2.25%	0.98%	+3.23%	0.879	+4.01%
model 5	3M	-0.90%	8.32%	+9.22%	0.879	+9.00%
model 5	6M	3.48%	19.09%	+15.61%	0.842	+14.86%
model 5 gated	1M	-2.25%	0.98%	+3.23%	0.879	+4.01%
model 5 gated	3M	-0.90%	8.32%	+9.22%	0.879	+9.00%
model 5 gated	6M	3.48%	19.09%	+15.61%	0.842	+14.86%
momentum only	1M	-2.25%	0.09%	+2.34%	0.855	+3.52%
momentum only	3M	-0.90%	4.45%	+5.35%	0.830	+9.79%
momentum only	6M	3.48%	10.94%	+7.46%	0.503	+12.08%

quality only	1M	-2.25%	-1.85%	+0.40%	0.915	+0.97%
quality only	3M	-0.90%	0.76%	+1.66%	0.636	+0.48%
quality only	6M	3.48%	8.14%	+4.66%	0.612	+2.96%
random ranking	1M	-2.25%	-1.58%	+0.67%	0.212	+0.02%
random ranking	3M	-0.90%	-0.89%	+0.01%	0.285	+0.61%
random ranking	6M	3.48%	2.84%	-0.64%	0.576	+2.80%
simple quality momentum	1M	-2.25%	-0.06%	+2.19%	0.855	+2.93%
simple quality momentum	3M	-0.90%	5.85%	+6.75%	0.903	+9.48%
simple quality momentum	6M	3.48%	13.53%	+10.05%	0.467	+12.84%
value only	1M	-2.25%	0.89%	+3.14%	0.467	+1.69%
value only	3M	-0.90%	8.46%	+9.36%	0.661	+5.31%
value only	6M	3.48%	65.41%	+61.93%	0.733	+21.80%

Turnover

One-way turnover of a top-20 portfolio per rebalance. 1.0 means the portfolio was fully replaced. High turnover is what converts a real gross edge into no net edge.

Strategy	Mean one-way turnover	Rebalances
equal weight universe	0.055	20
growth only	0.158	20
model 5	0.362	20
model 5 gated	0.362	20
momentum only	0.468	20
quality only	0.300	20
random ranking	0.958	20
simple quality momentum	0.398	20
value only	0.158	20

Fill coverage

Share of scored positions that produced a usable forward return. A dropped position is recorded with a reason rather than silently omitted; a thin horizon here weakens every statistic above it.

Horizon	Coverage	Positions	Main reasons dropped
1M	99.6%	129177	unadjustable_action 558, no_entry_price 9
3M	98.7%	129177	unadjustable_action 1620, no_entry_price 9
6M	97.5%	129177	unadjustable_action 3204, no_entry_price 9

How to read this

The random_ranking row is the calibration. It is a seeded random score, so its IC is what "no signal" measures on this universe and horizon. Any strategy whose IC is not clearly separated from it has not demonstrated anything.

equal_weight_universe scores every name identically, so its rank IC is undefined rather than zero. It is present as the return benchmark from p0.md §7B, not as a ranking.

Judge consistency before magnitude. A mean IC of 0.03 with 70% of periods positive is stronger evidence than a mean of 0.06 driven by two periods. Check the %+ and worst columns before the mean.

Net is the number that matters. If a gross edge disappears net of costs, the ranking may still be real but is not tradable at this position size and turnover.

Known limitations of this run

Restatement bias remains. No fundamentals are used here, so it does not bite yet — but it will once the XBRL panel lands, for any period falling back to current values rather than as-filed ones.

Delisting recovery is an assumption, not an observation. The bhavcopy does not record why a security stopped trading. Re-run with --delisting-strategy zero and last_close to bound how much the headline depends on it.

Period count is modest. Monthly rebalancing over roughly four years is enough to judge cross-sectional rank IC and its stability. It is not enough to claim regime robustness across a full cycle.

Overlapping horizons are not independent. A 6M horizon sampled monthly reuses most of its window, so any t-statistic on per-period ICs overstates significance and is reported as a rough guide only.

Cost assumptions are documented estimates. Brokerage defaults to zero (discount-broker delivery); half-spread and impact coefficient are modelled, not measured. Gross and net are shown separately for that reason.

Next step

Phase B: ingest NSE XBRL filings into a point-in-time fundamental panel keyed on filing timestamp, which unlocks the quality, growth and value blocks and makes a genuine Model 5.0 walk-forward possible. Until then the fundamental 70% of the score remains untested.